

A Level Mathematics 9709 · S1

Probability & Statistics 1

全套中英文详细学习手册

Detailed bilingual student manual · concept, derivation, worked examples
and exam method

本册共 8 个独立课时。每课按三页展开：第一页建立概念、推导方法并完成基础例题；第二页处理 Cambridge 风格应用、英文表达、评分点与易错辨析；第三页扩展考纲边界、变式推演与迁移任务。题目均为依据考纲原创的训练示例，不复制历年试题。

使用范围与版本 | Scope

0580 册依据 Cambridge IGCSE Mathematics 0580 (2025/2027) 的 Core/Extended 分层；9709 册依据 Cambridge International AS & A Level Mathematics 9709 (2026/2027)。M2 单独标为拓展/旧体系衔接材料，不作为现行独立考试组件。

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2. 排列组合 | Permutations and combinations
3. 概率规则 | Probability rules
4. 离散随机变量 | Discrete random variables
5. 二项分布 | Binomial distribution
6. 几何分布 | Geometric distribution
7. 正态分布 | Normal distribution
8. S1综合建模 | Statistics synthesis

第 1 课 | 数据表示 | Representation of data

Lesson 1 of 8 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

图表选择取决于数据类型；箱线图强调中位数与四分位，茎叶图保留原始值。

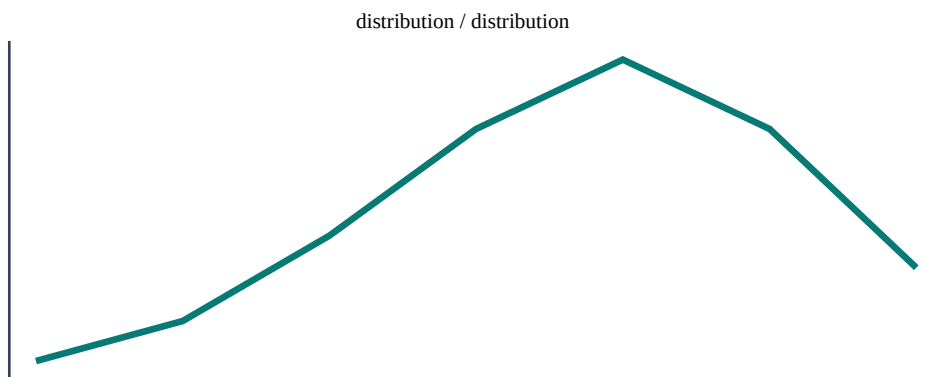
English core explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

为什么方法成立 | Reasoning behind the method

比较两组必须同时谈位置与离散，并联系情境。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

A组median 42,IQR 8; B组median 39,IQR 15, 比较。

A typical value higher and results more consistent because its median is larger and IQR smaller.

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 1 课（续） | 数据表示 | Representation of data

例题二：Cambridge 风格应用 | Cambridge-style application

Why is a histogram different from a bar chart?

Histogram bars touch and area represents frequency for continuous grouped data.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

mark。若前一步算错但方法仍一致，后续常可获得

follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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再推一步 | Extension and connection

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Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

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Detailed English explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

变式推演 | Variation and transfer

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第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：A组 median 42, IQR 8; B组 median 39, IQR 15, 比较。

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Cambridge 题型如何包装 | How the idea is examined

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第三个迁移任务 | Transfer task

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独立完成后对照：Histogram bars touch and area represents frequency for continuous grouped data.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 2 课 | 排列组合 | Permutations and combinations

Lesson 2 of 8 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

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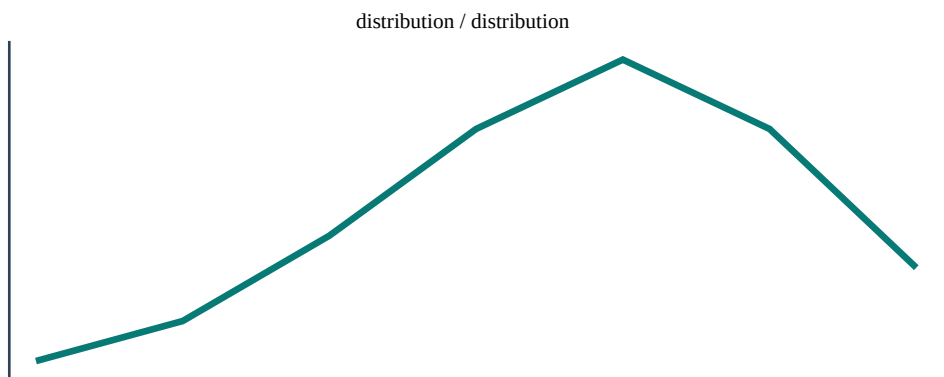
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为什么方法成立 | Reasoning behind the method

$nPr = n! / (n-r)!$; $nCr = n! / [r!(n-r)!]$ 。

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8人中选3人委员会有多少种？

$8C3=56$ 。

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第 2 课（续） | 排列组合 | Permutations and combinations

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How many arrangements of LEVEL?

$5! / (2!2!) = 30$ because L and E each repeat twice.

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Lesson 3 of 8 · Detailed bilingual learning unit

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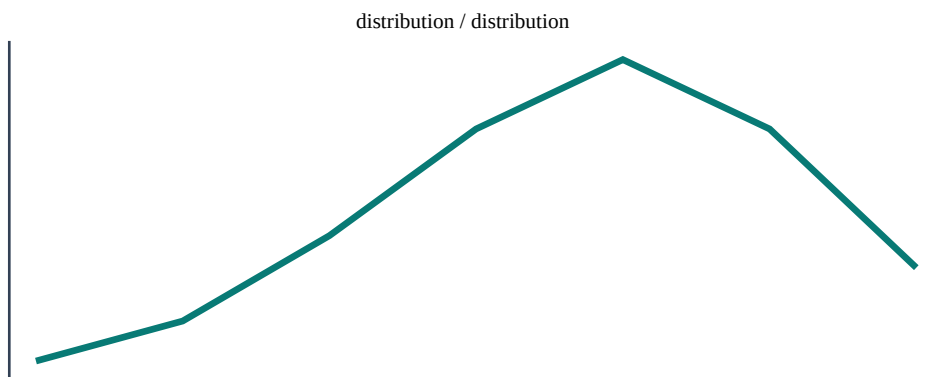
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为什么方法成立 | Reasoning behind the method

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)。$$

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例题一：从题意到完整解答 | Worked example 1

$P(A)=0.6, P(B)=0.5, P(A \cap B)=0.3$, 求 $P(A \cup B)$ 。

$$0.6+0.5-0.3=0.8。$$

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第 3 课（续） | 概率规则 | Probability rules

例题二：Cambridge 风格应用 | Cambridge-style application

Are A and B independent?

Yes, because $P(A)P(B)=0.6 \times 0.5=0.3=P(A \cap B)$.

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第三个迁移任务 | Transfer task

Are A and B independent?

独立完成后对照：Yes, because $P(A)P(B)=0.6 \times 0.5=0.3=P(A \cap B)$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 4 课 | 离散随机变量 | Discrete random variables

Lesson 4 of 8 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

概率分布列出X每个可能值及概率；期望是长期平均，不一定是可出现值。

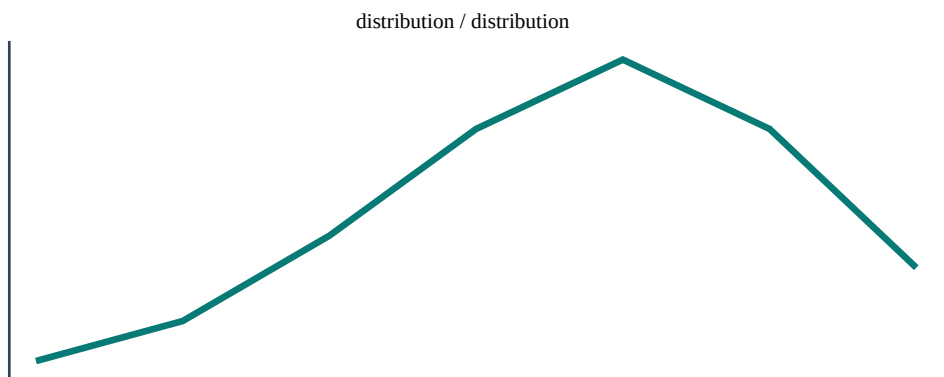
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为什么方法成立 | Reasoning behind the method

$$\sum p=1; E(X)=\sum xp; \text{Var}(X)=E(X)[E(X)].$$

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例题一：从题意到完整解答 | Worked example 1

X取0,1,2, 概率0.2,0.5,0.3, 求E(X)。

$$0+0.5+0.6=1.1。$$

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 4 课（续） | 离散随机变量 | Discrete random variables

例题二：Cambridge 风格应用 | Cambridge-style application

Find $\text{Var}(X)$.

$E(X)=0+0.5+1.2=1.7$; $\text{variance}=1.71.1=0.49$.

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Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

概率分布列出X每个可能值及概率；期望是长期平均，不一定是可出现值。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。 $\sum p=1$ ； $E(X)=\sum xp$ ； $Var(X)=E(X)[E(X)]$ 。

Detailed English explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：X取0,1,2，概率0.2,0.5,0.3，求 $E(X)$ 。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $0+0.5+0.6=1.1$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...” → “Therefore the required value is ...” . For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Find $Var(X)$.

独立完成后对照： $E(X)=0+0.5+1.2=1.7$; $variance=1.7 \cdot 1.1=0.49$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 5 课 | 二项分布 | Binomial distribution

Lesson 5 of 8 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

固定次数、两结果、独立、成功概率不变时 $X \sim B(n, p)$ 。

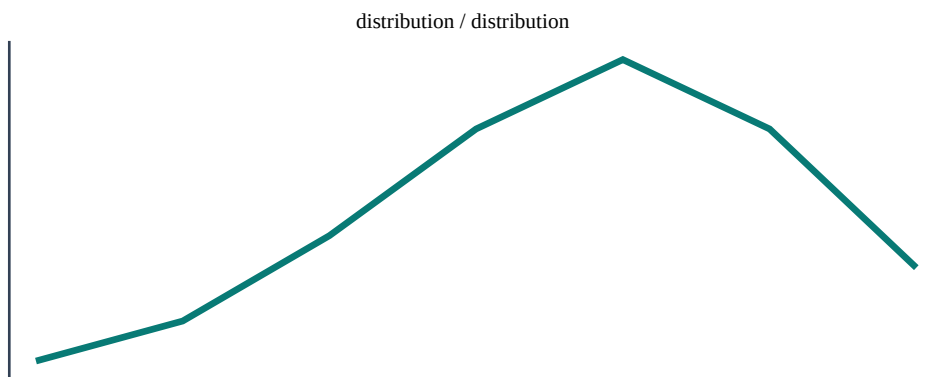
English core explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

为什么方法成立 | Reasoning behind the method

$P(X=r) = {}^nC_r p^r (1-p)^{(n-r)}$; at least 常用补集。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

$X \sim B(10, 0.2)$, 求 $P(X=2)$ 。

${}^{10}C_2 (0.2)^2 (0.8)^8 \approx 0.302$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 5 课（续） | 二项分布 | Binomial distribution

例题二：Cambridge 风格应用 | Cambridge-style application

Find $P(X \geq 1)$.

$$1P(X=0)=10.8^{10} \approx 0.893.$$

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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知识边界与相邻考点 | Scope and connections

固定次数、两结果、独立、成功概率不变时 $X \sim B(n, p)$ 。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。 $P(X=r) = {}^nC_r p^r (1-p)^{n-r}$; at least 常用补集。

Detailed English explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

变式推演 | Variation and transfer

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第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题： $X \sim B(10, 0.2)$ ，求 $P(X=2)$ 。

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Cambridge 题型如何包装 | How the idea is examined

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英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...” → “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Find $P(X \geq 1)$.

独立完成后对照： $1 - P(X=0) = 1 - 10 \cdot 0.8^9 \approx 0.893$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 6 课 | 几何分布 | Geometric distribution

Lesson 6 of 8 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

等待第一次成功的试验次数服从几何分布。

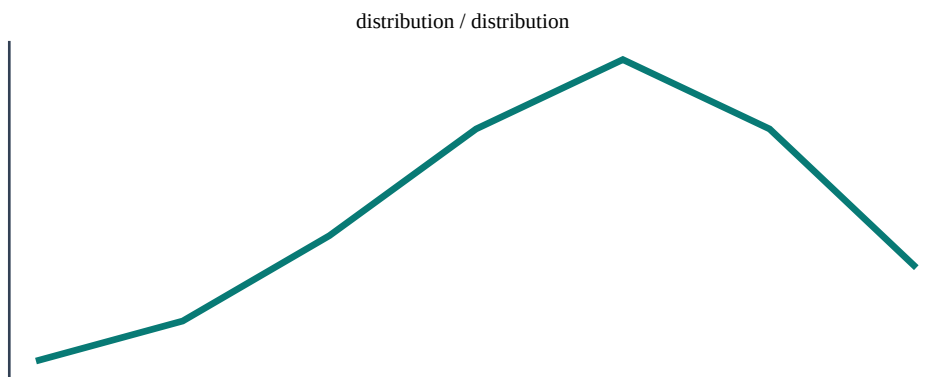
English core explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

为什么方法成立 | Reasoning behind the method

$$P(X=r)=(1p)^{(r1)}p; \quad P(X>r)=(1p)^r。$$

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
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4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

成功率0.3，首次成功在第4次概率。

$$0.7 \times 0.3=0.1029。$$

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 6 课（续） | 几何分布 | Geometric distribution

例题二：Cambridge 风格应用 | Cambridge-style application

Find probability first success occurs after trial 5.

$$0.7^5 \approx 0.1681.$$

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

等待第一次成功的试验次数服从几何分布。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。 $P(X=r)=(1p)^{(r-1)}p$; $P(X>r)=(1p)^r$ 。

Detailed English explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

变式推演 | Variation and transfer

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第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：成功率0.3，首次成功在第4次概率。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $0.7 \times 0.3=0.1029$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

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英文作答骨架 | Useful response frames

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第三个迁移任务 | Transfer task

Find probability first success occurs after trial 5.

独立完成后对照： $0.7^5 \approx 0.1681$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 7 课 | 正态分布 | Normal distribution

Lesson 7 of 8 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

正态分布由 μ 和 σ 决定，对称且总面积1；标准化 $Z=(X-\mu)/\sigma$ 。

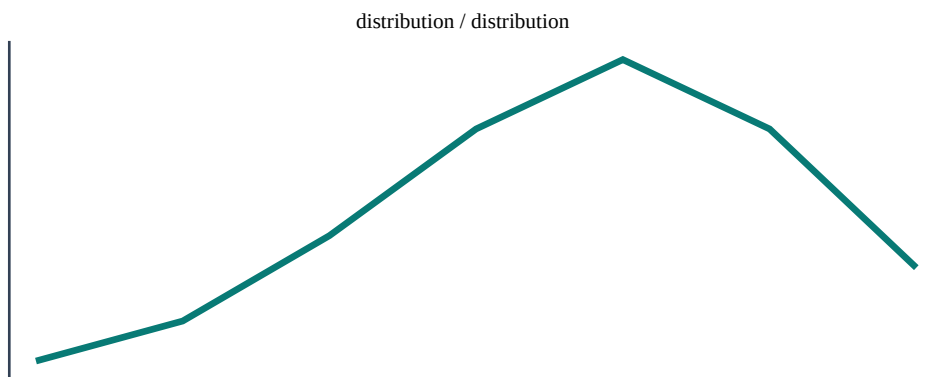
English core explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

为什么方法成立 | Reasoning behind the method

画草图、标均值与阴影，再用表/计算器；反求值时从概率找 z 。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

$X \sim N(50, 9)$ ，求 $P(X < 56)$ 。

$\sigma = 3$, $z = (56 - 50)/3 = 2$, $P \approx 0.9772$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 7 课（续） | 正态分布 | Normal distribution

例题二：Cambridge 风格应用 | Cambridge-style application

Find a with $P(X < a) = 0.1$.

$z \approx 1.2816$, so $a = 501.2816 \times 3 \approx 46.2$.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

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Detailed English explanation. Begin by defining the event or random variable. Check whether trials are independent, whether probability is constant, and whether the distribution is discrete or continuous. State parameters, show the probability statement, and interpret the result in context.

变式推演 | Variation and transfer

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第三道示范：审题、计算、复核 | Worked reasoning example

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先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $\sigma=3$ ， $z=(56-50)/3=2$ ， $P \approx 0.9772$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” $\rightarrow$ “Using ..., we obtain ...” $\rightarrow$ “Substituting and simplifying gives ...”
 $\rightarrow$ “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Find a with $P(X < a) = 0.1$.

独立完成后对照： $z \approx 1.2816$, so $a = 50 + 1.2816 \times 3 \approx 53.84$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 8 课 | S1综合建模 | Statistics synthesis

Lesson 8 of 8 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

先识别模型条件，再计算；最终概率必须在0到1之间。

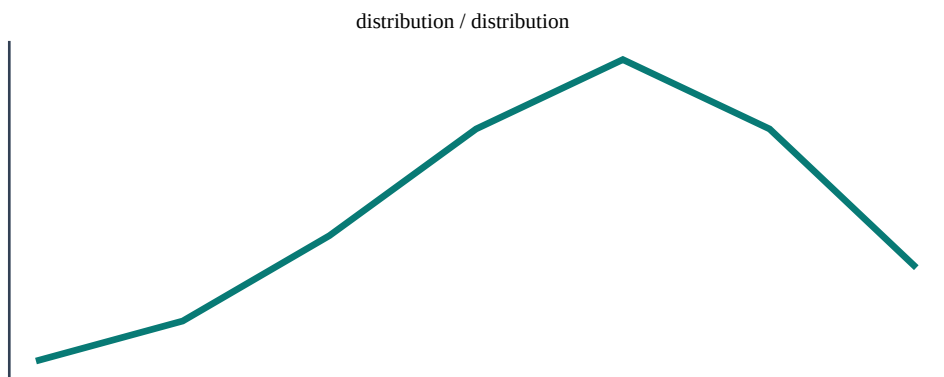
English core explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

为什么方法成立 | Reasoning behind the method

独立、固定 p 或正态近似的条件需从题干验证。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

何时不能用二项模型？

试验不独立、成功概率改变、次数不固定或结果不止成功/失败时。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 8 课（续） | S1综合建模 | Statistics synthesis

例题二：Cambridge 风格应用 | Cambridge-style application

Why should calculator notation be translated?

An answer needs the named distribution, parameters, event and probability, not only an unexplained decimal.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得

follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

先识别模型条件，再计算；最终概率必须在0到1之间。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。独立、固定 p 或正态近似的条件需从题干验证。

Detailed English explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：何时不能用二项模型？

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